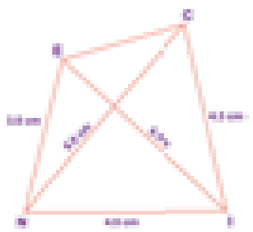
5.10.2 Construct a quadrilateral when its 3 sides and 2 diagonals are given

Example 5.22

Construct a quadrilateral *NICE* with *NI*=4.5 *cm*, *IC*= 4.3 *cm*, *NE*= 3.5 *cm*, *NC*= 5.5 *cm* and *IE* = 5 *cm*. Also find its area.

Solution:

Given:
NI = 4.5 *cm*, *IC*= 4.3 *cm*,
NE = 3.5 *cm* and two diagonals,
NC = 5.5 *cm* and *IE* = 5 *cm*



Rough diagram

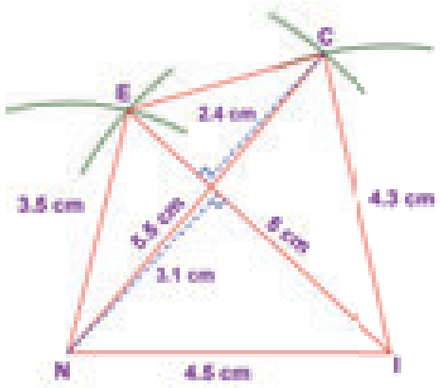


Fig. 5.41

Steps:

1. Draw a line segment *NI* = 4.5 *cm*.
2. With *N* and *I* as centres, draw arcs of radii 5.5 *cm* and 4.3 *cm* respectively and let them cut at *C*.
3. Join *NC* and *IC*.
4. With *N* and *I* as centres, draw arcs of radii 3.5 *cm* and 5 *cm* respectively and let them cut at *E*.
5. Join *NE*, *IE* and *CE* .
6. *NICE* is the required quadrilateral.

Calculation of Area:

Area of the quadrilateral *NICE* $= \frac{1}{2} \times d \times (h_1 + h_2)$ *sq. units*

$= \frac{1}{2} \times 5 \times (2.4 + 3.1)$

$= 2.5 \times 5.5 = 13.75\ cm^2$

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